

He-Ne Laser

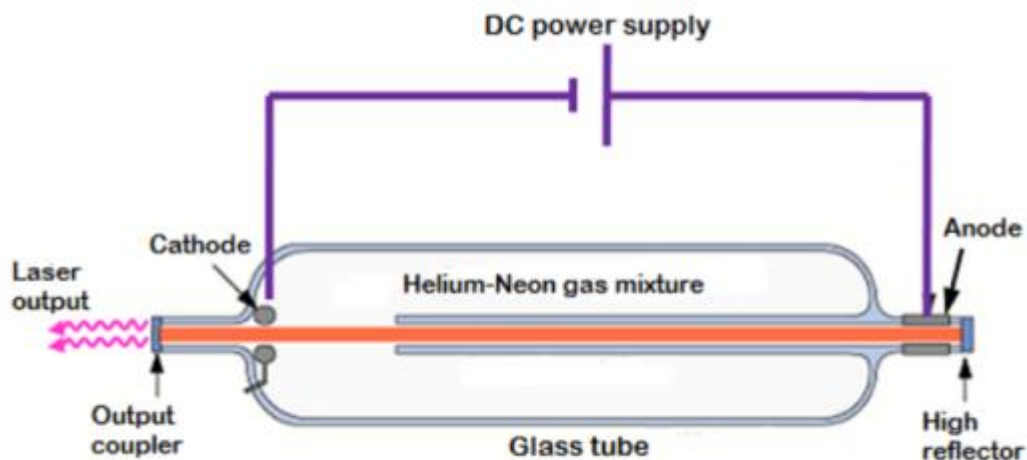
The Helium-Neon (He-Ne) laser is four level laser system as there are four energy levels involved. It is most popular visible light laser of red colour. The common wavelength of He-Ne laser is 632.8 nm. It is tunable from infrared to various visible light frequencies.

Active Medium: The Helium-Neon (He-Ne) laser is a kind of neutral atom gas laser. It consists of gas mixture containing 15% of Helium gas and 85% of Neon gas filled in a glass tube.

Pumping Mechanism: Pumping is done by a dc electrical discharge in a low-pressure discharge-tube. First, the He atom is excited. The Ne atom has an energy level very near to an energy level of He, therefore through kinetic interaction, energy is readily transferred from He to Ne; and the Ne atom emits the desired laser light.

Optical Resonator: A perfectly reflecting flat mirror is mounted at one end of the gas tube and a partially reflecting concave mirror is placed at the other end. The concave mirror focuses parallel light at the flat mirror and acts as a lens that transmits part of the light so that it emerges as a parallel beam.

Figure given below shows a schematic sketch of an He-Ne laser.



Helium-Neon laser is the most popular continuous wave (CW) laser. The energy levels of helium and neon, which are important for operation of the laser, are shown in diagram given below.

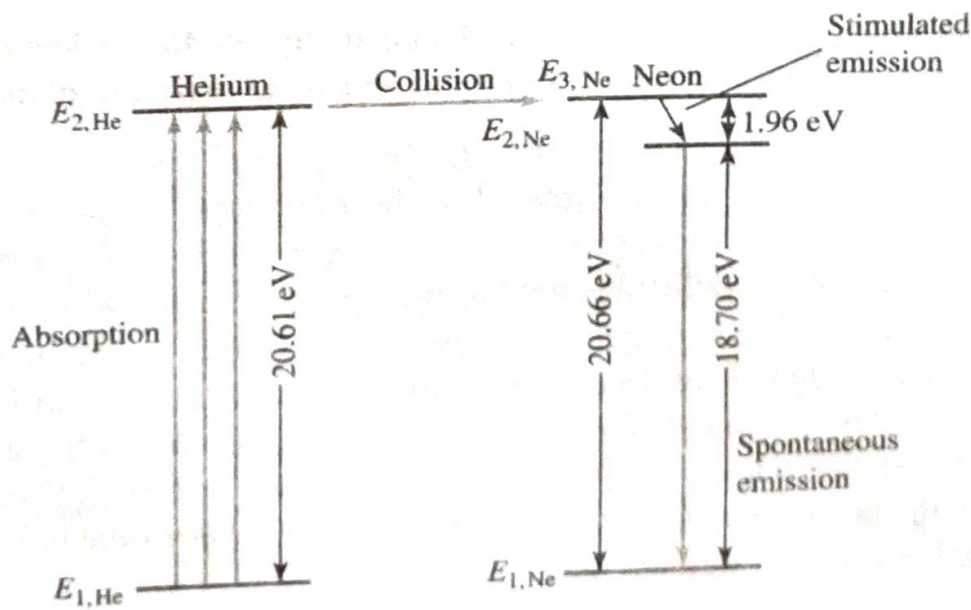


Fig. 5.16 Energy levels of the He-Ne laser

Working: Helium has an excited energy state $E_{2,He}$ at 20.61 eV above the ground state $E_{1,He}$. Only helium atoms are excited to state $E_{2,He}$ by an electric discharge. Neon has an excited state $E_{3,Ne}$ that is 20.66 eV above its ground state. This is very close to the energy state of Helium, i.e., just 0.05 eV above the first excited state of helium. The neon atoms are excited to the state $E_{3,Ne}$ by collision with the excited helium atoms. The kinetic energy of the helium atoms provides the extra 0.05 eV of energy needed to excite the neon atoms.

There is lower excited state of Neon $E_{2,Ne}$ at 18.70 eV above its ground level and 1.96 eV below the excited state $E_{3,Ne}$. Since state $E_{2,Ne}$ is normally unoccupied (metastable state), population inversion between states $E_{3,Ne}$ and $E_{2,Ne}$ is obtained immediately. The stimulated emission occurring between these states results in photons of energy 1.96 eV (632.8 nm), which produces a bright red light. After stimulated emission, the atoms in the state $E_{2,Ne}$ decay to the ground state by spontaneous emission.

The typical power of an He-Ne laser is below 50 mW, hence it is widely used in holography, barcode scanners, industrial measurements, optical fiber communication, scientific and laboratory work and in interferometers.